n8n

MASTERY ROADMAP

& COURSE OUTLINE

For Developers & Automation Professionals • Beginner to Top 1%

Designed by Raju J • AI Workflow Engineer • 2025–2026

10	60+	25+	∞
Modules	Hours Content	Real Projects	Skills Gained

Course Overview

This is NOT another n8n basics course. This roadmap is engineered specifically for developers and automation professionals who want to go beyond YouTube tutorials and build production-grade AI automation systems used by the top 1% of n8n practitioners.

WHY THIS ROADMAP IS DIFFERENT
Most courses teach n8n as a tool. This course teaches n8n as a platform — covering what competitors miss: production deployment, Docker/Kubernetes scaling, custom node development, MCP integration, AI agent orchestration, security hardening, and multi-client routing systems.

What You Will Master

- Design and deploy end-to-end AI-powered automation workflows
- Build RAG chatbots, multi-agent systems, and AI routing engines
- Integrate LLMs (OpenAI, Gemini, Claude, Ollama) with real business systems
- Deploy n8n to production using Docker, VPS, and Kubernetes
- Build and publish custom n8n community nodes
- Implement MCP (Model Context Protocol) for AI agent tool exposure
- Scale workflows with Redis queues, parallel execution, and worker pools
- Secure n8n for enterprise use with RBAC, secrets management, and auditing
- Build AI Voice Agents, WhatsApp bots, and multi-channel automation
- Architect multi-client SaaS automation platforms

Prerequisites

- Basic understanding of REST APIs and JSON
- Familiarity with any programming language (Python/JS preferred)
- Basic command line / terminal usage
- A Google account and access to OpenAI or Gemini API (free tier sufficient)

Learning Roadmap — 4 Phases

Phase	Focus Area	Modules	Duration
Phase 1	Foundation & Core Concepts	Modules 1–2	~8 hours
Phase 2	AI Integration & Workflow Design	Modules 3–5	~18 hours
Phase 3	Production, Scaling & Security	Modules 6–8	~20 hours
Phase 4	Advanced AI Systems & Monetization	Modules 9–10	~14 hours

PHASE 1 — FOUNDATION

Modules 1–2 • ~8 Hours • Core Platform Mastery

MODULE 1	n8n Core Architecture & Setup	~4 hrs
Topics Covered	▸ n8n vs Zapier vs Make — when and why n8n wins ▸ Installation: Local, Cloud, Docker, VPS ▸ Interface deep-dive: Canvas, nodes, credentials ▸ Triggers: Webhook, Schedule, Manual, App triggers ▸ Data flow: JSON structure, expressions, $node references ▸ Credentials management & OAuth2 setup ▸ Built-in nodes: HTTP Request, Code, Set, IF, Switch, Merge ▸ Error handling: Try/Catch, Continue on Fail, Error Workflows ▸ Execution logs, debugging, and testing strategies ▸ n8n 2.0 breaking changes & migration guide	**Hands-on Projects**
		✦ **GitHub → Notion sync workflow** ✦ **Scheduled data scraper + Google Sheets logger** ✦ **Multi-branch conditional router**

MODULE 2	API Mastery & Data Engineering	~4 hrs
Topics Covered	▸ HTTP Request node: Auth, pagination, rate limiting ▸ REST API patterns: GET/POST/PUT/DELETE/PATCH ▸ Webhook: inbound, response, security headers ▸ JSON manipulation: jmespath, $json, $item() ▸ Array operations: split, batch, aggregate, filter ▸ Data transformation: Code node with JavaScript ▸ Working with files: CSV, PDF, images in workflows ▸ Database nodes: MySQL, PostgreSQL, MongoDB, Supabase ▸ GraphQL integration ▸ API authentication: Bearer, API key, OAuth2, Basic	**Hands-on Projects**
		✦ **Multi-source data aggregation pipeline**

		✦ **REST API wrapper workflow (expose as webhook)** ✦ **Automated database sync system**

PHASE 2 — AI INTEGRATION

Modules 3–5 • ~18 Hours • LLMs, Agents & RAG

MODULE 3	LLM Integration & Prompt Engineering	~6 hrs
Topics Covered	▸ OpenAI node: Chat, Completions, Embeddings, Images ▸ Gemini, Claude, Mistral, Ollama integration ▸ Prompt engineering for automation contexts ▸ Structured output: JSON mode, function calling ▸ Token management, cost optimization, fallback chains ▸ AI Chain vs AI Agent — when to use each ▸ Memory nodes: Buffer, Summary, Window, Vector ▸ System prompts, few-shot examples, chain-of-thought ▸ AI content moderation with Guardrails node ▸ Multi-LLM routing and load balancing	**Hands-on Projects**
		✦ **Intelligent email classifier & auto-responder** ✦ **AI-powered content generation pipeline** ✦ **Multi-model comparison workflow**

MODULE 4	RAG Systems & Vector Databases	~6 hrs
Topics Covered	▸ RAG architecture: Retrieval, Augmentation, Generation ▸ Vector stores: Supabase, Pinecone, Qdrant, Weaviate ▸ Embedding generation: OpenAI, Gemini, local models ▸ Document loaders: PDF, URL, Google Drive, Notion ▸ Chunking strategies: fixed, semantic, recursive ▸ Semantic search: similarity thresholds, reranking ▸ Hybrid search: vector + keyword (BM25) ▸ RAG evaluation: faithfulness, relevancy, context recall ▸ Streaming responses in n8n chat workflows ▸ Multi-tenant knowledge base isolation	**Hands-on Projects**
		✦ **Document Q&A chatbot (PDF + Supabase)** ✦ **Company knowledge**

		base RAG system ✦ **200+ doc semantic search engine**

MODULE 5	AI Agents & Multi-Agent Orchestration	~6 hrs
Topics Covered	▸ AI Agent node: tools, memory, planning loop ▸ Tool nodes: Calculator, Wikipedia, Web Search, Custom ▸ ReAct pattern: Reason → Act → Observe loop ▸ Sub-workflow agents: decompose complex tasks ▸ Multi-agent teams: Orchestrator + Specialist agents ▸ Agent-to-agent communication via webhooks ▸ Human-in-the-loop: approval gates, Slack/email confirms ▸ Agentic workflows: autonomous task completion ▸ Browser automation agents (Playwright integration) ▸ Agent failure handling and retry strategies	**Hands-on Projects**
		✦ **Research agent (web search + summarize + report)** ✦ **Multi-agent lead qualification system** ✦ **Autonomous customer support agent team**

PHASE 3 — PRODUCTION & SCALE

Modules 6–8 • ~20 Hours • Enterprise-Grade Deployment

MODULE 6	Production Deployment & Infrastructure	~7 hrs
Topics Covered	▸ Docker Compose: n8n + PostgreSQL + Redis + Nginx ▸ Environment variables, .env files, secrets injection ▸ Queue mode: Redis-backed worker pool setup ▸ Horizontal scaling: multiple worker containers ▸ Kubernetes deployment with Helm charts ▸ Auto-scaling worker pods based on queue depth ▸ PostgreSQL vs SQLite — production database choice ▸ Reverse proxy: Nginx/Traefik, SSL/TLS with Certbot ▸ Zero-downtime deployments with rolling updates ▸ n8n 2.0 hardening: secure-by-default execution	**Hands-on Projects**
		✦ **Full Docker Compose production stack** ✦ **Multi-worker queue setup (5+ concurrent workflows)** ✦ **Kubernetes deploy with auto-scaling**

MODULE 7	Security, Monitoring & DevOps	~7 hrs
Topics Covered	▸ n8n RBAC: Admin, Editor, Viewer + Custom Project Roles ▸ API key management, credential encryption at rest ▸ Workflow-level secrets with n8n Variables ▸ Audit logging: execution history, user actions ▸ Guardrails node: PII scrubbing, prompt injection defense ▸ Source control: Git integration for workflow versioning ▸ Prometheus + Grafana monitoring for n8n metrics ▸ Alerting: execution failure, queue depth, latency spikes ▸ Automated backups: PostgreSQL dumps + S3 ▸ OWASP top 10 applied to n8n webhook endpoints	**Hands-on Projects**
		✦ **Full monitoring dashboard (Prometheus + Grafana)** ✦ **Git-based workflow**

		CI/CD pipeline ✦ **Security audit & hardening checklist**

MODULE 8	Custom Nodes & Platform Extension	~6 hrs
Topics Covered	▸ n8n node architecture: INodeType interface ▸ Scaffolding a custom node with n8n-node-dev CLI ▸ Credential classes: OAuth2, API Key, HTTP Auth ▸ Node properties: UI inputs, resources, operations ▸ Execute method: IExecuteFunctions, $getInputData() ▸ Error handling in custom nodes ▸ Testing custom nodes locally ▸ Publishing to npm as community node ▸ Versioned nodes: maintaining backward compatibility ▸ Node UI: dynamic fields, routing, display conditions	Hands-on Projects
		✦ **Custom CRM integration node** ✦ **Published npm community package** ✦ **Versioned node with migration strategy**

PHASE 4 — ADVANCED AI SYSTEMS

Modules 9–10 • ~14 Hours • MCP, Voice & SaaS

MODULE 9	MCP Integration & Agentic AI Platform	~7 hrs
Topics Covered	▸ MCP (Model Context Protocol) — architecture deep-dive ▸ n8n as MCP Server: exposing workflows as AI tools ▸ n8n as MCP Client: consuming external MCP servers ▸ Claude Desktop + n8n MCP integration ▸ MCP OAuth authentication flow ▸ Tool discovery, schema definition, parameter validation ▸ AI Voice Agents: n8n + Retell AI / VAPI / ElevenLabs ▸ Voice workflows: STT → LLM → TTS → telephony ▸ WhatsApp, Telegram, Discord AI bot orchestration ▸ Multi-channel unified AI agent hub	Hands-on Projects
		✦ **n8n MCP server exposing 10+ tools to Claude** ✦ **AI Voice Agent for appointment booking** ✦ **Unified multi-channel AI bot (WhatsApp + Telegram)**

MODULE 10	Multi-Client SaaS & Monetization	~7 hrs
Topics Covered	▸ Multi-tenant workflow architecture patterns ▸ Client isolation: separate credentials, namespaces ▸ Webhook URL routing: per-client dynamic endpoints ▸ n8n Embed: white-label automation for SaaS products ▸ Usage metering: execution counts, API call tracking ▸ Billing integration: Stripe webhooks + n8n automation ▸ Template library: reusable workflow marketplace ▸ Client onboarding automation pipeline ▸ SLA monitoring: uptime, error rate, response time alerts ▸ Building an AI Automation Agency with n8n	Hands-on Projects
		✦ **10-client routing system with**

		isolated contexts ✦ **White-label automation platform (n8n Embed)** ✦ **Full SaaS: billing + onboarding + monitoring**

Capstone Projects

Complete at least 2 of the following capstone projects to demonstrate mastery:

Capstone A — Enterprise AI Automation Suite

Build a full multi-agent AI system that handles lead capture via WhatsApp, qualifies leads with an AI agent, syncs to CRM, sends personalized emails, and generates weekly analytics reports — fully autonomous.

Capstone B — Production RAG Platform

Deploy a multi-tenant RAG chatbot platform with isolated vector stores per client, document ingestion pipelines, real-time streaming responses, usage analytics, and a Stripe billing integration.

Capstone C — AI Voice Agent System

Build an AI voice agent using n8n + VAPI/Retell AI that handles inbound calls, books appointments via Cal.com, sends SMS confirmations, and logs all interactions to a CRM — exposed as MCP tools.

Capstone D — n8n SaaS Automation Agency

Launch a white-label automation platform using n8n Embed with client onboarding, isolated workflow environments, usage metering, Stripe billing, and a dashboard showing ROI metrics per client.

Full Technology Stack

Category	Tools & Technologies
Automation	n8n (self-hosted & cloud), n8n 2.0, Queue Mode, Worker Pools
AI / LLMs	OpenAI GPT-4o, Google Gemini, Anthropic Claude, Ollama (local), Mistral
Vector DBs	Supabase (pgvector), Pinecone, Qdrant, Weaviate, Chroma
Deployment	Docker, Docker Compose, Kubernetes (Helm), VPS (Ubuntu), n8n Cloud
Databases	PostgreSQL, MySQL, MongoDB, Redis (queue), SQLite (dev only)
Monitoring	Prometheus, Grafana, n8n Audit Logs, Sentry (error tracking)
Voice / Bots	Retell AI, VAPI, ElevenLabs, Twilio, WhatsApp API, Telegram Bot
Security	Nginx, Certbot (SSL), Vault, RBAC, Guardrails node, Git versioning
MCP	n8n-nodes-mcp, Claude Desktop, MCP Server Trigger, MCP Client Tool
Integrations	Stripe, Cal.com, Google Workspace, Notion, Airtable, HubSpot, Slack

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